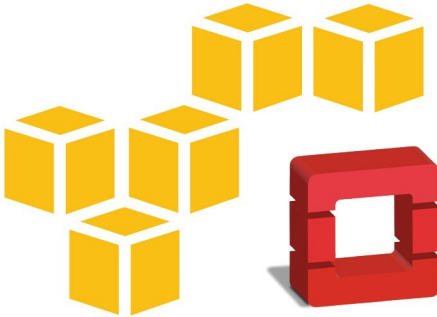


A Guide to Running OpenStack on AWS

OpenStack is a cloud operating system that controls vast computing resources through a data centre, while AWS (Amazon Web services) offers reliable, scalable and inexpensive cloud computing services. The installation of OpenStack on AWS is an instance of Cloud-on-a-Cloud.



This article will guide you on installing OpenStack on top of AWS EC2. Installing OpenStack on a nested hypervisor environment is not a big deal when a QEMU emulator is used for launching virtual machines inside the virtual machine (VM). However, unlike the usual nested hypervisor set-up, installing OpenStack on AWS EC2 instances has a few restrictions on the networking side, for the OpenStack set-up to work properly. This article outlines the limitations and the solutions to run OpenStack on top of the AWS EC2 VM.

Limitations

The AWS environment will allow the packets to flow in their network only when the MAC address is known or registered in the AWS network environment. Also, the MAC address and the IP address are tightly mapped. So, the AWS environment will not allow packet flow if the MAC address registered for the given IP address is different.

You may wonder whether the above restrictions will impact the OpenStack set-up on AWS EC2.

Yes, they certainly will! While configuring Neutron networking, we create a virtual bridge (say, *br-ex*) for the provider network, where all the VM's traffic will

reach the Internet via the external bridge, followed by the actual physical NIC (say, *eth1*). In that case, we usually configure the external interface (NIC) with a special type of configuration, as given below.

The provider interface uses a special configuration without an IP address assigned to it. Configure the second interface as the provider interface. Replace *INTERFACE_NAME* with the actual interface name, for example, *eth1* or *ens224*.

Next, edit the */etc/network/interfaces* file to contain the following:

```
# The provider network interface
auto INTERFACE_NAME
iface INTERFACE_NAME inet manual
up ip link set dev $IFACE up
down ip link set dev $IFACE down
```

You can go to <http://docs.openstack.org/mitaka/install-guide-ubuntu/environment-networking-controller.html> for more details.

Due to this special type of interface configuration, the restriction in OpenStack will hit OpenStack networking. In the mainstream OpenStack set-up, the above-mentioned provider